

FTML practical session 7

3 juin 2026

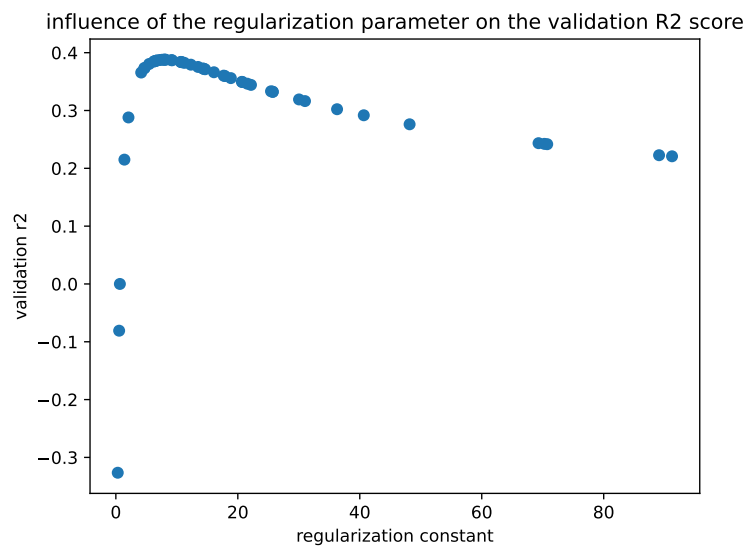


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1 HYPERPARAMETER TUNING METHODS

In this exercise we summarize the most common hyperparameter tuning methods and use **optuna** as a bayesian optimization framework.

1.1 Grid search and random search

Grid search is the most basic approach to hyperparameter tuning : choose a list of values for each and iterate through a grid containing all possible combinations.

https://scikit-learn.org/stable/modules/grid_search.html#

See also :

- random search
- successive halving.

1.1.1 Issues

Grid search might be suboptimal in the sense that all hyperparameter combinations will be tried, although many of them might be irrelevant. Also, the size of the grid is exponential in the number of parameters.

1.1.2 Question

Why can we often not run a gradient-based optimization algorithm to look for better HP values?

1.2 Bayesian optimization

More modern approaches use smarter optimization methods to tune the hyperparameters. For instance, **Bayesian optimization**.

https://en.wikipedia.org/wiki/Bayesian_optimization

1.2.1 Frameworks

Several frameworks exist to perform Bayesian optimization over HPs values, and automate the search.

skopt <https://scikit-optimize.github.io/stable/>

optuna <https://optuna.org/>

With these methods, we can specify ranges for scalar HP values, instead of grids of values.

1.2.2 Optuna and optuna dashboard

Optuna creates "studies", in which HP sets are tested in a sequence. A set of HPs is tested in a "trial".

<https://optuna.readthedocs.io/en/stable/reference/study.html>

Some interesting optuna aspects to consider :

- some trials can be "pruned" : they are considered as not interesting by optuna, and are discarded before the optimization is finished.

- it is possible to start a study, stop the program, and resume the study later, with the previous trials still in the study database.
- you can analyze the results of a study by processing the study object with other libraries like pandas, seaborn, matplotlib, etc.

With optuna, you can display and analyze the influence of the HP with a dashboard that opens in your browser.

<https://github.com/optuna/optuna-dashboard>

Perform HP optimization with optuna over a problem of your choice. Visualize the results in the optuna dashboard

Suggestion : if you want, you can use **main_optuna_template.py** and the data contained in **data/**, in **exercise_3_optuna/** folder (for apply a ridge regression on this dataset), but you are encouraged to try also in a different problem, for instance with scikit's toy datasets or any dataset of your choice.

2 INFLUENCE ON DATA SCALING ON THE CONVERGENCE OF SGD

We would like to perform a binary classification on the following 3 datasets (figures 1, 2, 3). Each one has a different difficulty for a linear classifier, such as a SVM. We also note that the scales are different on the x and y axes.

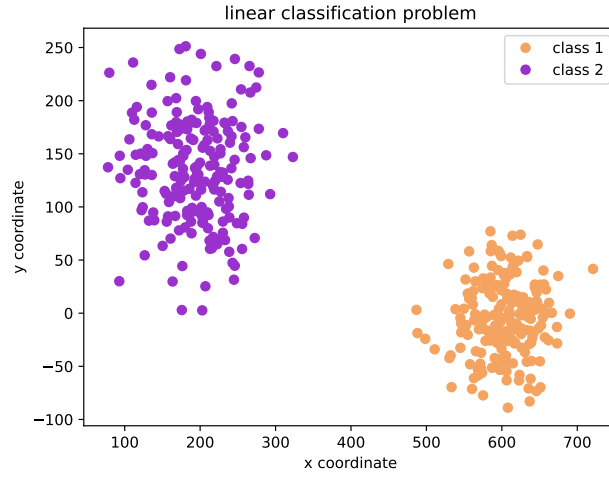


FIGURE 1 – Dataset 1

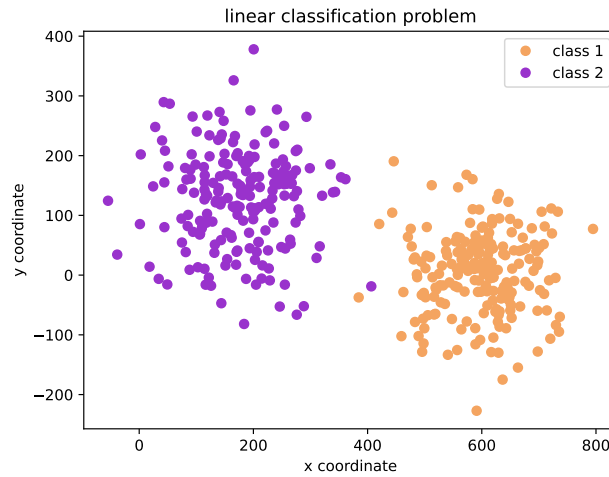


FIGURE 2 – Dataset 2

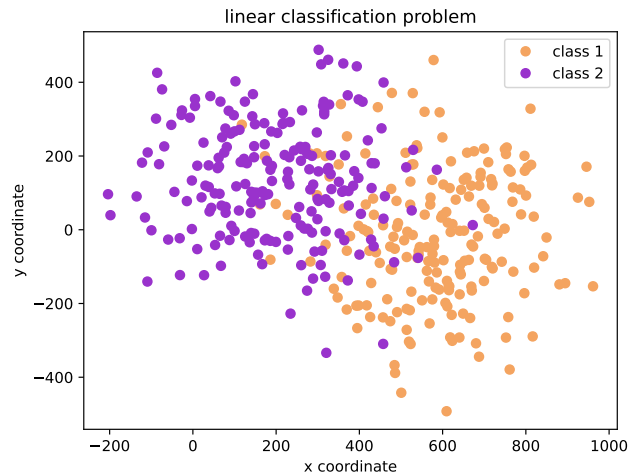


FIGURE 3 – Dataset 3

The data are located in `exercice_3_svm_sgd/data/`.

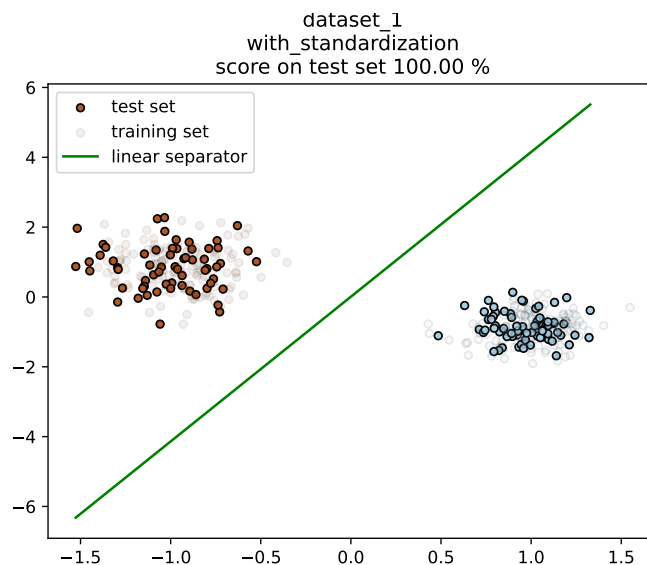
Data standardization consists in transforming the data so that each feature (each column) is centered (zero mean) and has a variance equal to 1. It is experimentally often noticed that algorithms trained by SGD give a better performance (generalization error) when the data are standardized.

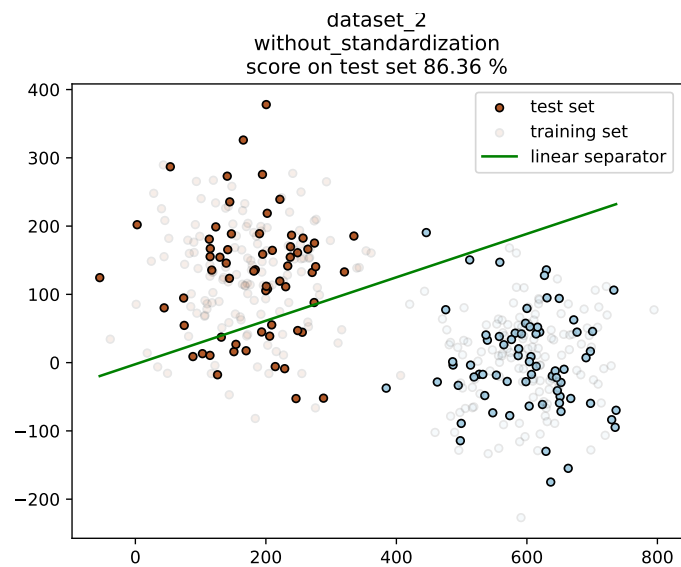
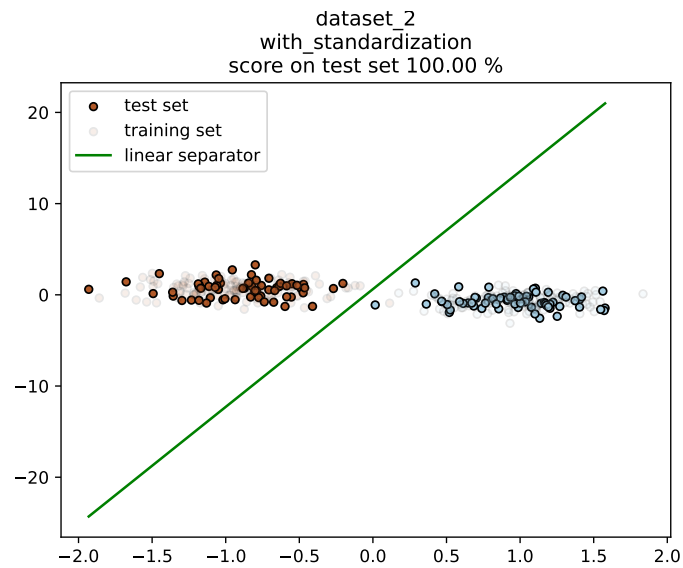
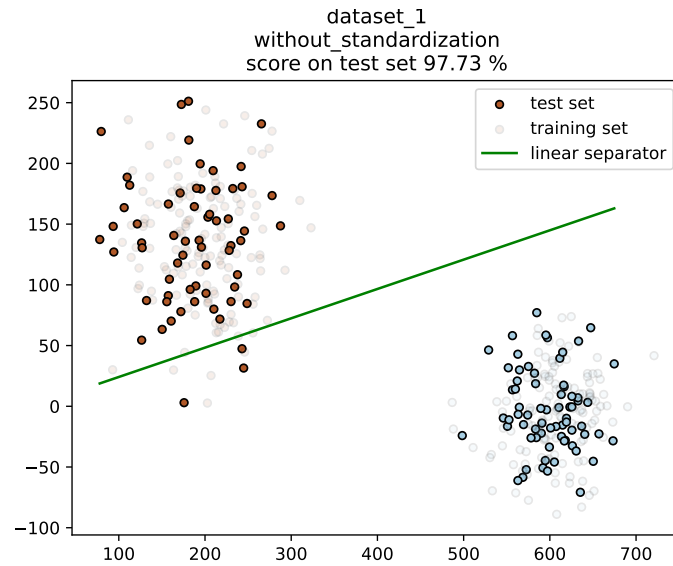
<https://scikit-learn.org/stable/modules/preprocessing.html#preprocessing-scaler>.

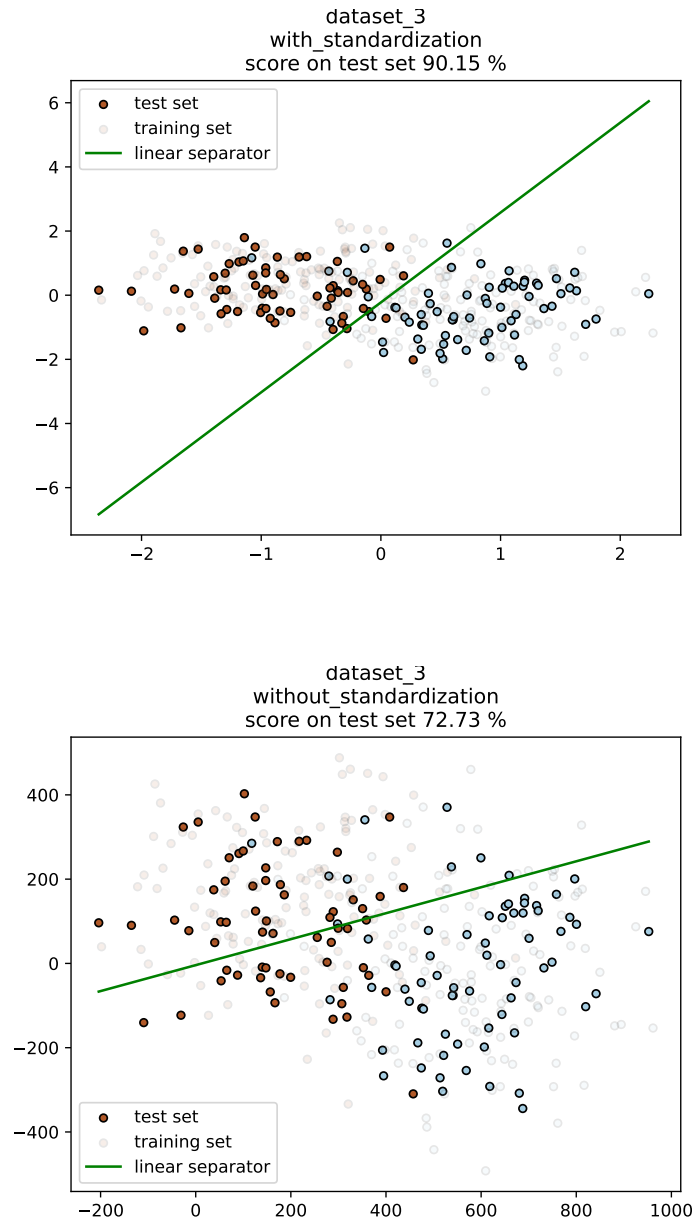
Perform a linear classification on these data, optimized by SGD, and compare quality of the results with and without preprocessing the data by standardizing them. You may use scikit-learn to do it, in particular :

- https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.SGDClassifier.html. By default, this class trains a linear SVM (no kernel).
- <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>
- <https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html>

You should obtain results like the following figures.







In this example, we saw that data standardization may give an improved performance, on a linear SVM trained by SGD. You can perform the same study on different datasets such as the toy datasets from scikit or other datasets.